# Ian W. McMurry

imcmurry3@gatech.edu | ian.mcmurry01@gmail.com |

*Research interests: Bayesian hierarchical modeling, statistical machine learning, and representation learning*

## Education

**Georgia Institute of Technology** — Atlanta, USA / Online

*Master of Science in Analytics; GPA: 4.00/4.00* — *Aug. 2024 – Expected Dec. 2026*

- Relevant coursework: Bayesian Statistics (in progress), Statistical Modeling and Regression Analysis, Computational Data Analysis, Simulation and Modeling for Engineering and Science, Applied Natural Language Processing.

**Emory University** — Atlanta, USA

*Bachelor of Business Administration; Finance and Information Systems; GPA: 3.77/4.00* — *Aug. 2019 – May 2023*

## Publications

**McMurry, Ian W.** "Quantifying Social Sentiment in Hostels Using A Domain-Specific Transformer Pipeline." — 2026

- In *Proceedings of the 15th Workshop on Computational Approaches to Subjectivity, Sentiment & Social Media Analysis (WASSA 2026)*, Association for Computational Linguistics, pp. 24–36. DOI: 10.18653/v1/2026.wassa-1.3.
- Developed a cross-encoder, pseudo-labeling, and domain-adapted bi-encoder pipeline on 162,840 hostel reviews; achieved F1 = 0.826 on held-out human-labeled data versus 0.671 for generic embeddings and 0.774 for zero-shot GPT-4o-mini.

## Research Experience

**Hierarchical Bayesian Modeling of Armenian Housing Rental Yields** — May 2026 – Present

*Independent Research Project* — *Remote*

- Developed a multivariate hierarchical Bayesian model in PyMC to recover composition-adjusted rental-yield posteriors from unpaired rent and sale listings, linking district effects through a learned bivariate-normal hierarchy.
- Fit the model with NUTS to 8,863 listings across 36 Armenian districts; obtained 91.3%/90.9% empirical coverage of nominal 90% posterior predictive intervals for rent/sale prices with zero divergent transitions.

**Hostel Review Representation Learning** — July 2024 – Present

*Independent Research Project* — *Remote*

- Built and maintained a corpus of 160,000+ hostel reviews across 70+ global cities and developed task-specific representation-learning and classification workflows for subjective experiential attributes.
- Extended the WASSA socialness pipeline to property-level benchmarking and additional guest-experience signals, including cleanliness, calmness, and review sentiment.
- Collaborating with Mad Monkey, an international hostel chain, to evaluate operational applications of review-derived experience metrics for property benchmarking and guest-experience analysis.

## Selected Presentations

**McMurry, Ian W.** "Quantifying Hostel Social Atmosphere with Domain-Specific NLP." — Accepted, Oct. 2026

- Invited for virtual poster presentation at the Georgia Tech 2026 OMS Analytics Annual Conference.

## Professional Experience

**U.S. Marine Corps Officer — Supply & Logistics** — July 2023 – Present

*United States Marine Corps* — *Okinawa, Japan*

- Led a 19-person logistics team across three supply accounts and later served as the principal Korea-theater logistics and contracting representative across four major joint planning cycles.
- Designed documentation and supply warehouse workflow controls that improved 30-day accountability from 60% to 92% and increased special-project parts shipped by 71%.

**Analytics Engineer Consultant** — June 2025 – Present

*S10 Consulting* — *Remote*

- Designed a multi-stage solicitation ingestion and insight-generation platform used by 5+ recurring clients to transform unstructured procurement notices into structured, searchable data.
- Integrated LLM-assisted extraction workflows to generate 11 structured attributes per solicitation and architected backend schemas supporting a scalable analytics product.

**Operations Data Analyst Intern** — Nov. 2022 – Nov. 2023

*TOCA Football* — *Remote*

- Analyzed over $1.3M in sales data across 14 locations and created KPI dashboards and Python/SQL-powered reporting workflows for 240+ trainers, saving 15+ hours per week.

**Teaching Assistant, Business Analytics** — Jan. 2023 – May 2023

*Emory University* — *Atlanta, USA*

- Supported 250+ students in analytics topics including regression, advanced Excel, macros, and scenario analysis; held weekly office hours and graded applied coursework.